Manu Nagendra Nalla

Backend Software Engineer | Go, Kafka, Distributed Systems, Data Migration
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PROFESSIONAL SUMMARY

Backend engineer with 6+ years building **distributed systems in Go (Golang)**, Kafka event pipelines and large-scale data migrations. At Kroger's Customer Platform: core engineer on a **1.22 billion-record production migration** at 99.99% accuracy, and creator of **CP Krovis**, a Kafka tool (Go, React, TypeScript) used by **40+ developers** across 4 teams. Holds an MS in Computer Science.

TECHNICAL SKILLS

Languages: Go (Golang), Python, TypeScript, JavaScript, SQL, Java, Bash
Backend and Messaging: Apache Kafka, Avro, Schema Registry, REST APIs, OpenAPI/Swagger, WebSockets, JWT, LDAP, Event-Driven Architecture, Event Sourcing, Microservices, Domain-Driven Design, OAuth, Dead Letter Queues
Databases: YugabyteDB, PostgreSQL, Redis, MongoDB, Cassandra, DuckDB, Parquet, PL/SQL
Cloud and DevOps: Kubernetes, Docker, Rancher, Harness, Spinnaker, GitHub Actions, CI/CD, Azure, Akeyless, Disaster Recovery
Observability and Frontend: Dynatrace, Grafana, Structured Logging, React, Vite, Vitest
Practices: Agile, Scrum, Test-Driven Development, DevOps, Technical Design Documentation

PROFESSIONAL EXPERIENCE

Software Engineer II | 05/2024 - Present
Ascendum Solutions (Client: Kroger), Cincinnati, OH

Customer Platform data migration and synchronization (Thesaurus team)

- Core engineer on the Customer Platform migration, delivering through every quarterly run from 24Q2 to close-out; owned the validation, error-analysis and fix-script streams for the final production run of about **1.22 billion records** across 12 entity types, migrated at **99.99% accuracy** with every residual delta on groups and alt IDs reconciled to zero.
- Matched about **211K card and digital-ID mismatches** between the legacy source and the migrated data against the migration's pre-processing error logs, tracing **99.98%** to documented source-data causes and proposing fixes per category, and contributed to triage of about 8.1M errors from the April 2025 production run.
- Cut the time to identify mismatches between source and migrated systems by about **80%** (from about a day with awk scripts to 1 to 2 hours) by loading database extracts into Parquet files and comparing them with **DuckDB** SQL scripts.
- Designed and documented (technical design document) a reusable **Go data-fix framework** (fetchers for three source systems, analyzer, executor CLI, JSON Merge Patch library) that replaced one-off scripts, and used it to repair altID, householdID, shopperID, email and missing-card mismatches in production.
- Built a Go random-data generator that creates millions of synthetic records matching CG, EC and CP schemas; used in every migration run since 24Q3, it caught **15+ critical transformation bugs** before production.
- Built a Failed Events API (dead-letter-queue-as-a-service) covering about **915K DLQ messages a month**, and an error-handling package that cut debugging time by about **73%** (2.5 hours to 40 minutes).
- Tuned forward-sync event consumption by load-testing transformer and loader counts against CPU allocation, raising throughput from about 4K to about 14K events per minute (**3.5x**).
- Wrote code for the Kafka sync services that handle **60.6M production requests a month** at a **0.0012% error rate** and 7.8 ms p95, feeding Customer Platform back ends that serve 28.6 billion calls a month.

CP Krovis: self-initiated Kafka explorer, built and owned end to end

- Built CP Krovis, now a production app used by **40+ developers across 4 engineering teams** to browse, search, stream and publish Kafka messages without CLI access, replacing one-off consumers and platform tickets; ran a walkthrough session for all four teams, and it is referenced in the disaster recovery runbooks for finding Kafka offsets by timestamp during region failover.
- Unified 12 Kafka environments behind one interface with Avro schema-registry decoding, multi-partition time-range search, live WebSocket streaming and JSON diff; secured with LDAP, JWT and audit logging, and covered by **180 automated tests** (115 Go, 65 Vitest) with CI/CD on GitHub Actions and Docker.

Customer Platform services (Phalcons team)

- Owned and released the Accounts, Groups, Search, Entities History, Orchestrator and Shim services: **14 production deployments across 25+ services** from July to September 2026 under a formal release-governance process.
- Worked on the Phalcons microservices (Accounts, Groups, Search, Entities History, Orchestrator and Shim), organized around business domains (domain-driven design), in an event-driven, event-sourced architecture on Kafka, with OAuth-based authentication.
- Built disaster recovery for eight Kafka consumers and DLQ processors, including Central-to-East failover runbooks, and validated CP Search at about **2,995 GET requests per second** (99.8% of target, 9.94 ms p95) and 585 POST requests per second (22.6 ms p95) against 150 ms and 300 ms SLAs.

Instructional Assistant | 08/2023 - 05/2024
California State University, Long Beach, Long Beach, CA

- Assisted the professor in running an algorithms course of 100+ students, increasing student pass rates by **20%** through targeted tutoring sessions and open office hours.
- Built an auto-grader handling 100+ submissions a week for fair evaluation of student algorithms, reducing grading time from **3 days to 10 minutes**.

Senior Software Development Engineer | 04/2022 - 12/2022
ZopSmart (Client: Kroger), Bengaluru, India

- Architected and maintained a data synchronization application using Kafka, Go and YugabyteDB, enabling data exchange between legacy and modern systems.
- Led a **10-member team** replacing a legacy enterprise customer data system, increasing migration efficiency by **50%**.
- Engineered an application that transformed and loaded over **450 million records** from legacy to modern systems, and automated migration procedures with a custom script.
- Promoted from SDE I to Senior SDE in a double promotion after 2 years.

Software Development Engineer I | 03/2020 - 03/2022
ZopSmart (Client: Kroger), Bengaluru, India

- Led an ETL-based migration project from inception to deployment; centralized customer data management, saving an estimated **$2.3M annually**.
- Centralized PII data, avoiding CCPA penalties of up to **$7,500 per incident**.
- Built projects from scratch, created proofs of concept, set up CI/CD, and wrote unit and integration tests.

SDE | 11/2019 - 03/2020
Accenture, India

- Supported Pfizer's PL/SQL systems: debugged stored procedures and triggers, optimized code for performance, and enhanced modules to meet business requirements.

PROJECTS

CP Krovis (2026): Internal Kafka message explorer built in Go, React and TypeScript and used by 40+ developers (details under Kroger experience).
recon (2026): Open-source Go CLI that reconciles two JSONL datasets and groups every difference by cause, with sample IDs, JSON output and CI-friendly exit codes (standard library only, with tests); github.com/manunag/recon
Data Loader (2022): Go application that loads data between databases with configurable filters and transformations (CSV, JSON, SQL, Postgres, Cassandra, Kafka).

EDUCATION

Master of Science in Computer Science | 01/2023 - 12/2024
California State University, Long Beach, Long Beach, CA
Bachelor of Engineering in Computer Science | 07/2015 - 05/2019
Saveetha School of Engineering, Chennai, India

Relevant coursework: Advanced Algorithms, Advanced Artificial Intelligence, Machine Vision, Pattern Recognition.

CERTIFICATIONS AND ACHIEVEMENTS

- Certification: Anthropic, Claude Code in Action (February 2026), credential ID nfpjfjyaymza
- Hosted tech sessions at ZopSmart; 1st place, Tame the Code, SAMVID, Saveetha University.